

Clustering and cell type identification

Code Club

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Overview

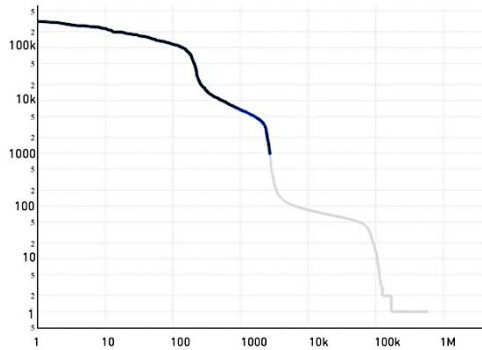
- 1) Choosing number of dimensions (principal components)
- 2) Choosing clustering resolution
- 3) Naming clusters
- 4) Clustering with multiple replicates
- 5) Choices of batch correction
- 6) Is anything real??

Questions

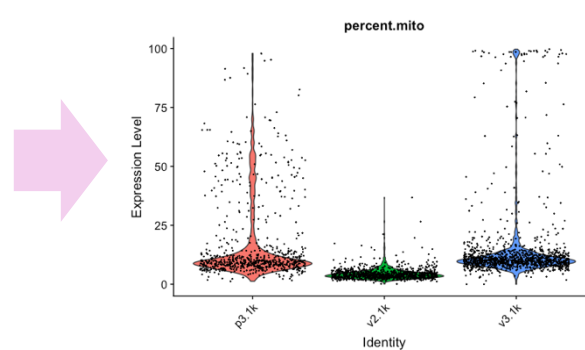
- 1) How arbitrary are the choices we make in single-cell RNA-seq analysis?
- 2) How confidently can we draw biological conclusions?

Overview of single-cell RNA-seq analysis

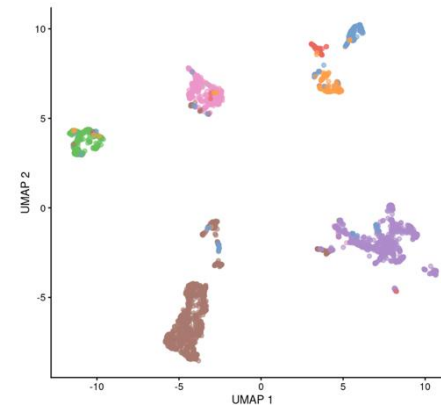
Identification of cells



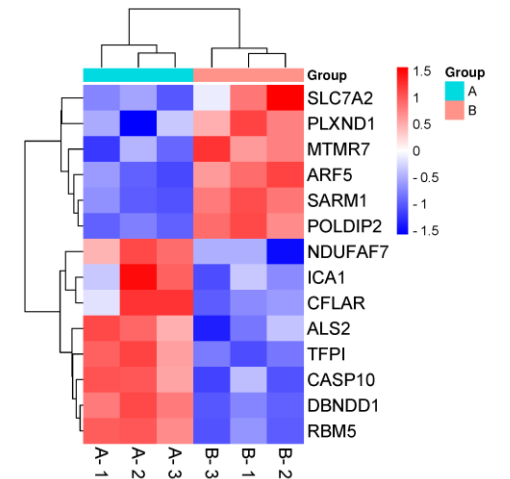
Quality filtering



Clustering and cell type identification



Downstream analysis



What is a Seurat object?

The Seurat object contains information about each cell, e.g.:

- Which sample it came from
- Total UMI counts per cell
- QC info like percent reads from mitochondrial genes
- Cell types
- PCA and UMAP reductions

seurat_data@

meta.data	{seurat_data}
assays	
active.assay	
active.ident	
graphs	
neighbors	
reductions	
images	
project.name	

seurat_data@reductions\$

pca
rna.umap
transcripts_pca
tx.umap

seurat_data\$

orig.ident
nCount_RNA
nFeature_RNA
nCount_TRANSCRIPTS
nFeature_TRANSCRIPTS
percent.mt
cell_qc_sum
cell_qc_detected
cell_qc_subsets_Mito_sum

What is a Seurat object?

The Seurat object contains the raw UMI counts per cell per gene:

```
> seurat_data@assays$RNA$counts
```

Each column is a cell



Iapp	482	185	776	22	898	286	22	1264	999	1314	18	899	22	598	32	25	813
Iars	.	.	.	.	.	1	.	1	.	1	.	1	.	.	1	.	.
Iars2	.	1	1	.	.	3	.	1	.	1	.	.	.	2	.	.	.
Iba57	.	1	.	.	.	1	.	.	.	.	.	.	.	.	.	.	.
Ibtk	2	.	.	.	2	.	.	2	.	1	.	.	.	.	.	.	.
Ica1	.	1	3	1	5	3	.	.	.	.	.	.	1	.	1	.	2

Each row is a gene



Numbers are UMI counts (per gene per cell)



What is a Seurat object?

After cell type identification, it will also contain cell type information per cell:

```
> seurat_data$RNA_barcode_celltype
```

```
AAACCTGAGTGTACTC AAACCTGCACATGTGT AAACCTGCACCTCGTT AAACCTGGTAAGGATT AAACCTGGTAGCCTCG  
      "beta"           "delta"           "beta"           "alpha"           "delta"  
AAAGCAAGTTGCCTCT AAAGCAATCGCATGGC AAAGTAGGTAAGTAGT AAAGTAGGTTACGTCA AAAGTAGTCTACTTAC  
      "beta"           "beta"           "beta"           "beta"           "delta"
```

← Cell barcodes (one barcode per cell)

← Cell type

Clustering

Why?

In scRNA-seq, each cell has thousands of genes measured, creating a **high-dimensional dataset**.

- Clustering helps us **group cells with similar expression pattern**.
- Usually, these groups correspond to **cell types, cell states, or subpopulations**.

How?

The first step of clustering is **Principal Component Analysis (PCA)** which is a linear dimensionality reduction.

- You're choosing how much of the variation in the dataset you want to keep for downstream analysis.
(variation = differences in gene expression patterns between cells)
- **Goal:** Keep the meaningful variation. Enough variation to capture the biological structure.
- **If you pick too few dimensions:**
 - You might miss biologically relevant variation.
- **If you pick too many dimensions:**
 - You might introduce noise and “overcluster” (split homogeneous populations artificially).

Principal Component Analysis (PCA)

Reduce dimensionality

Principal component analysis (PCA) takes a **high-dimensional dataset** (thousands of genes per cell) and transforms it into a **smaller set of new axes** called **principal components (PCs)**.

Each PC is a **weighted combination of all genes**.

PCs are **ranked by the amount of variation they explain**:

- **PC1** explains the largest variation.
- **PC2** explains the second largest, and so on.

Dimensions of Seurat gene expression matrix:

```
> dim(seurat_data)
[1] 22029 4814
```

of genes

of cells

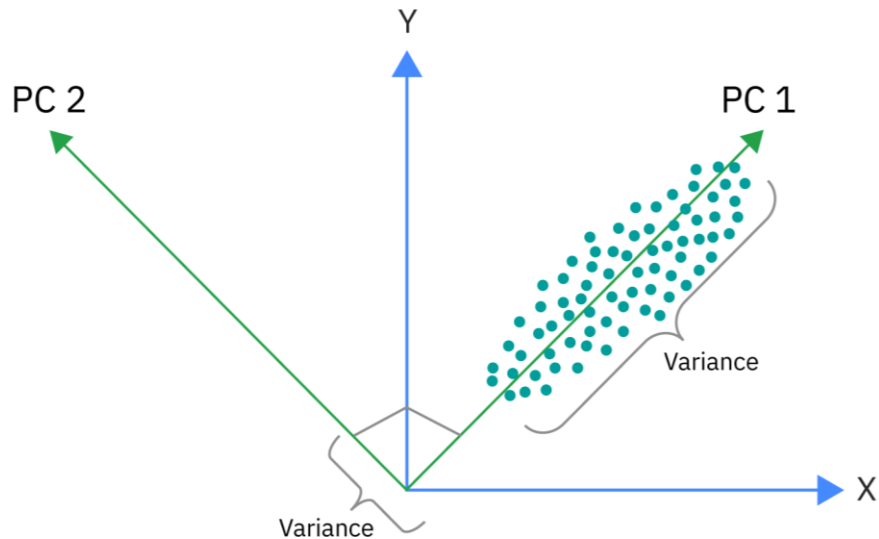
What does PCA do?

Imagine each gene is a dimension (axis).
Each cell is a point.

If you measure 20,000 genes, each cell is a point
in a 20,000-dimensional space.

PCA finds new axes (principal components)
pointing in directions with the most variation.

They explain most of the variation in much fewer
axes = they reduce dimensionality.



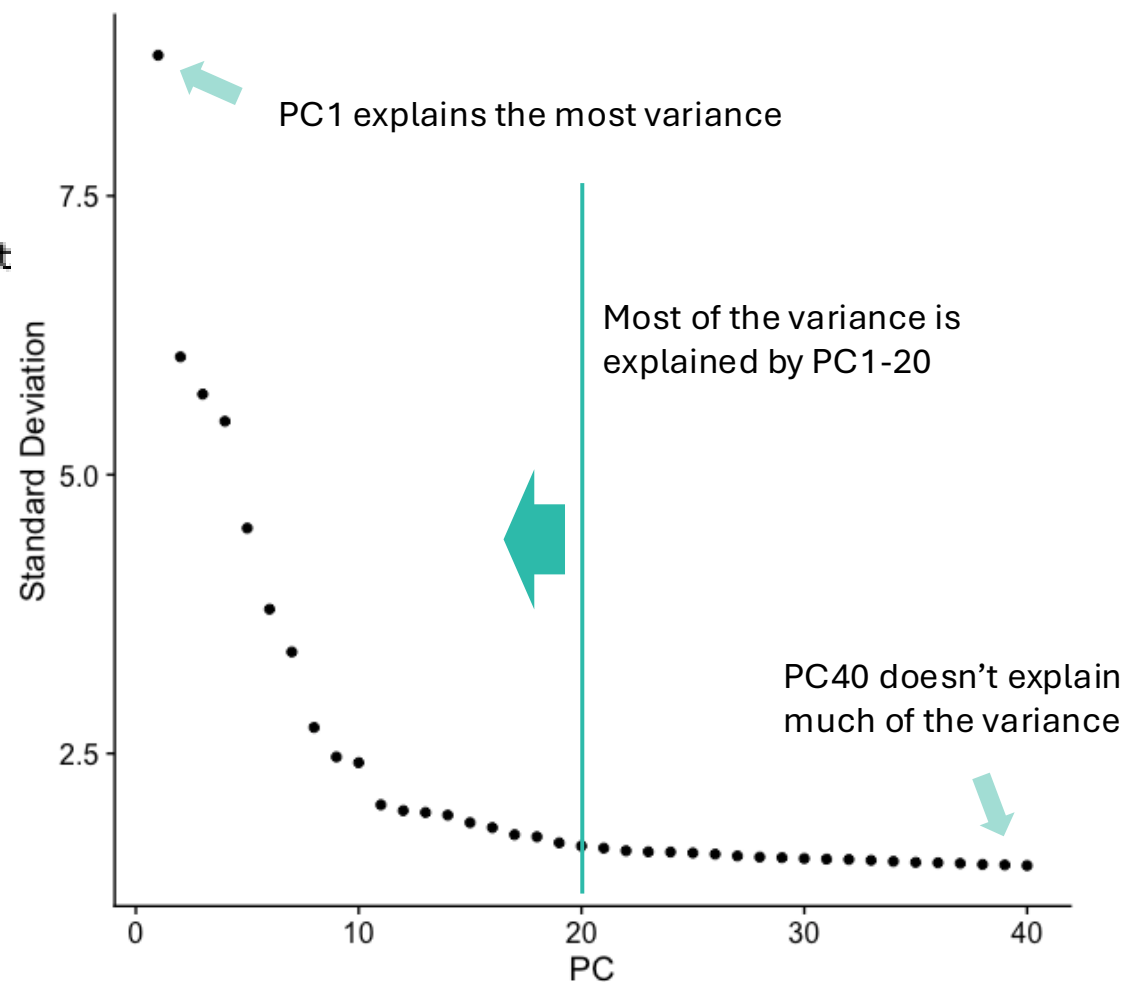
Principal Component Analysis (PCA)

```
# Load Seurat object:
seurat_data <- readRDS(file.path(save_dir, "islet_rep1.rds"))

# Run PCA:
seurat_data <- RunPCA(seurat_data,
                      features = VariableFeatures(object = seurat_data))

# Make Elbow plot:
ElbowPlot(object = seurat_data, ndims = 40)
```

The **Elbow Plot** shows how much variance each PC explains:

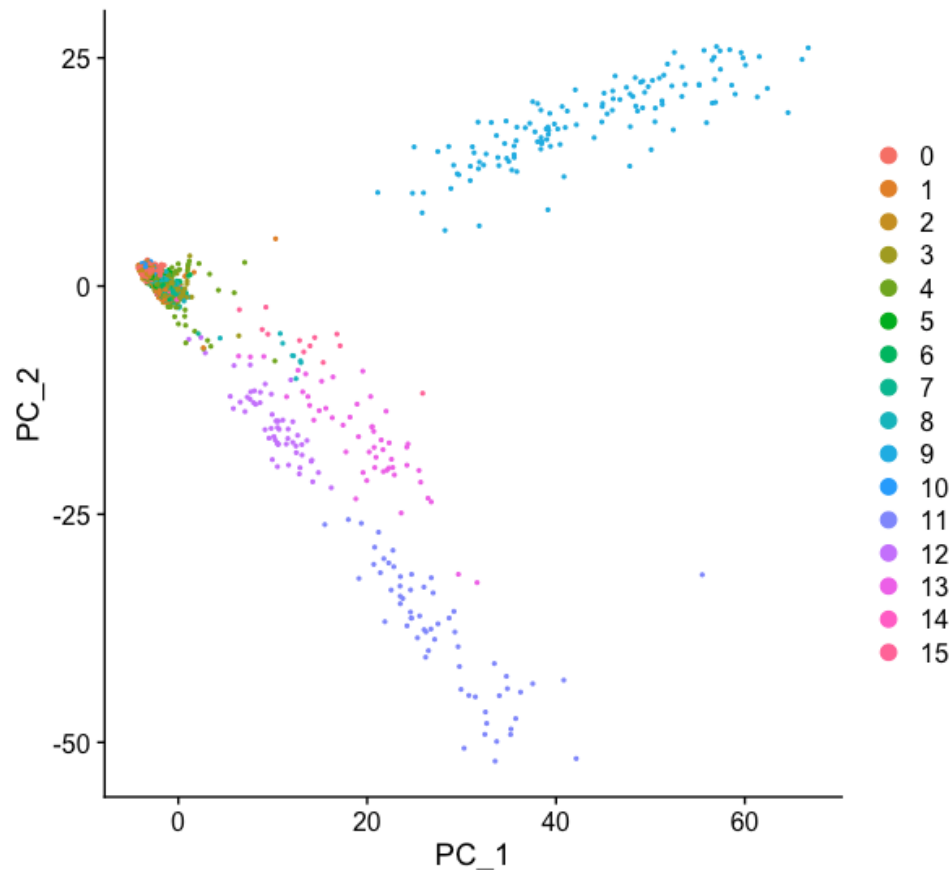


Principal Component Analysis (PCA)

What do the principal components look like?

```
# Make PCA scatter plot:
```

```
DimPlot(seurat_data, reduction = "pca", dims = c(1, 2))
```



Each axis is a **linear combination of genes**
(they're straight directions through the high dimensional space)

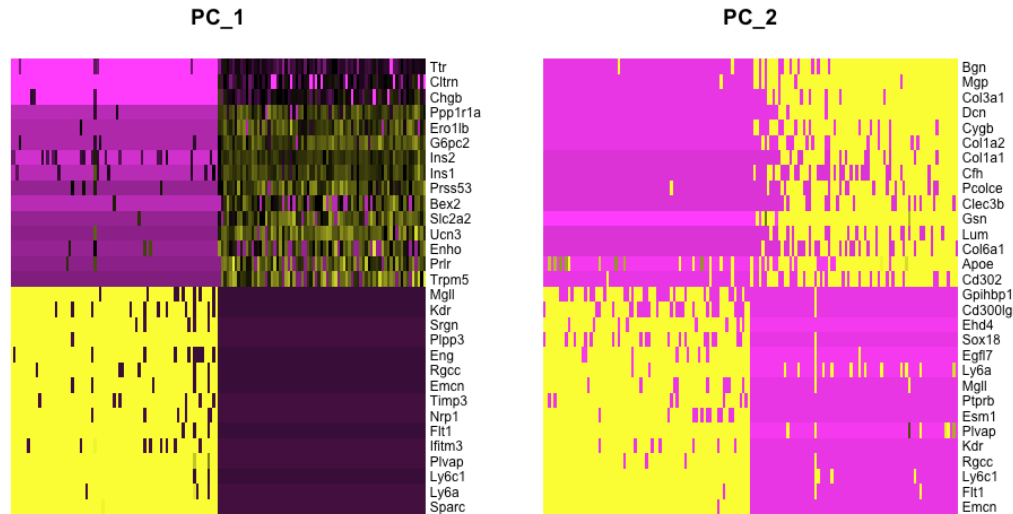
Cells that are **close together** in the plot have **similar patterns of expression** in the genes driving PC1 and PC2.

This is unlike the UMAP plot!

Principal Component Analysis (PCA)

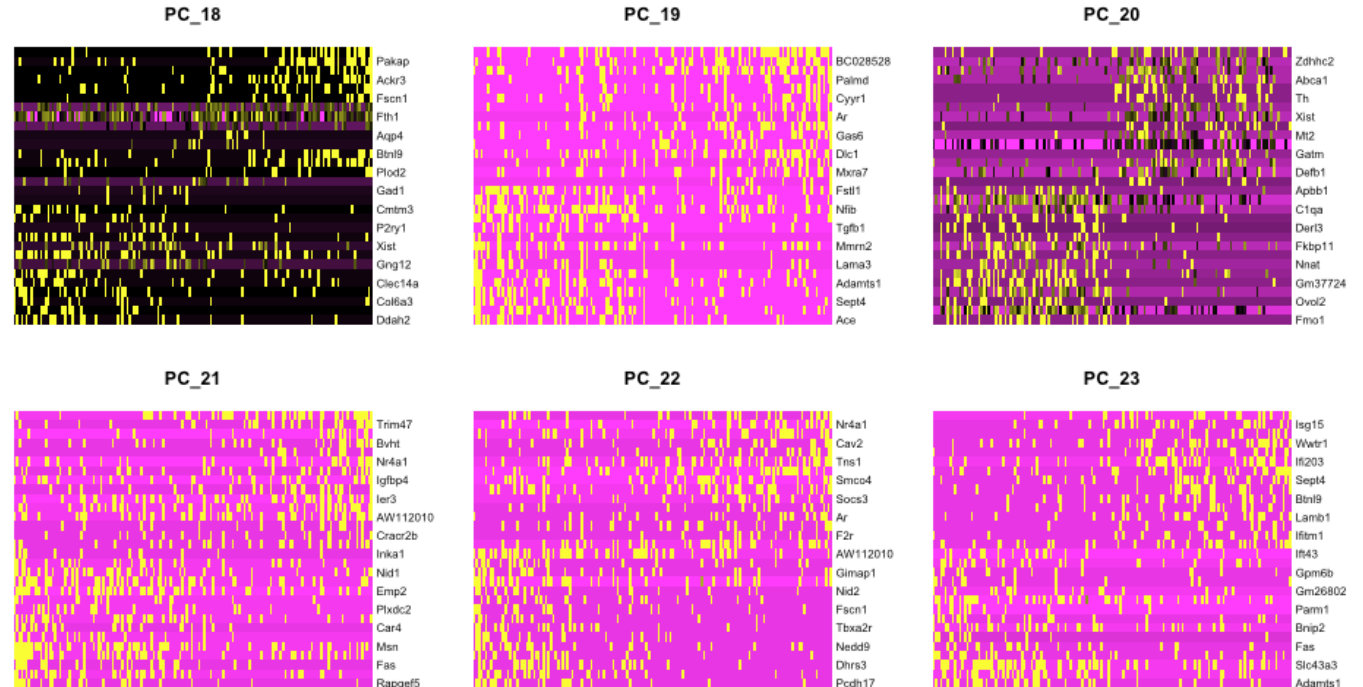
You can also use heat maps of genes positively or negatively associated with the PC to determine number of PCs to use:

```
# Make heatmap of genes contributing to PCs:  
DimHeatmap(seurat_data, dims = 1:2, cells = 150, balanced = TRUE)
```



The association starts to disappear at PC20:

```
# Make heatmap of genes contributing to PCs:  
DimHeatmap(seurat_data, dims = 18:23, cells = 150, balanced = TRUE)
```



Principal Component Analysis (PCA)

Signs you've picked too few PCs:

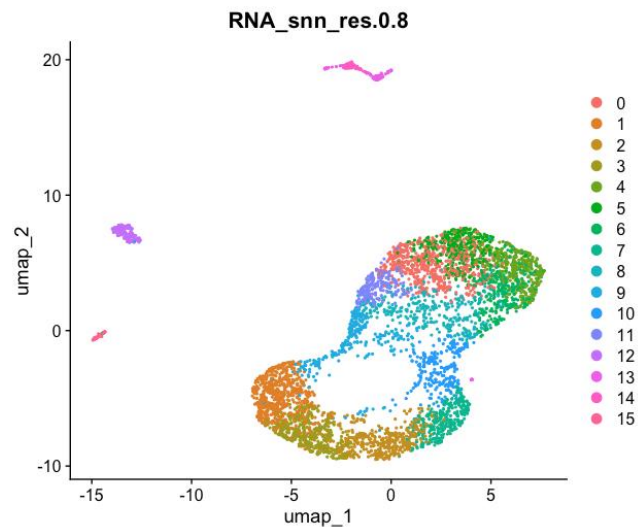
- Broad, merged clusters where known cell types aren't separated.
- UMAP looks like a blob with little internal structure

Signs you've picked too many PCs:

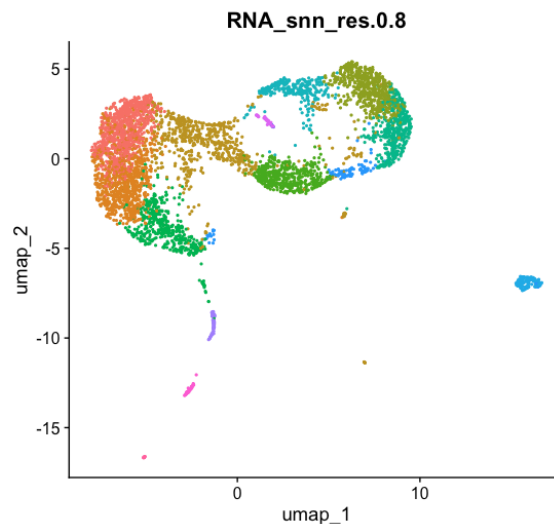
- Clusters are fragmented and tiny
- No clear biological meaning in the splits

Principal Component Analysis (PCA)

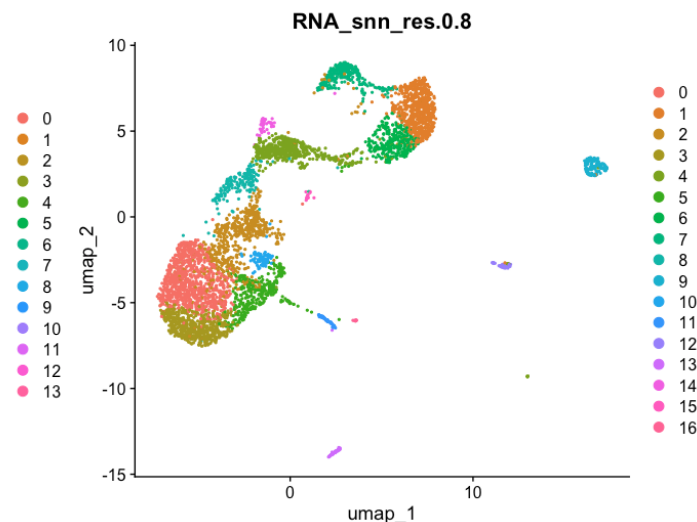
Dims = 5



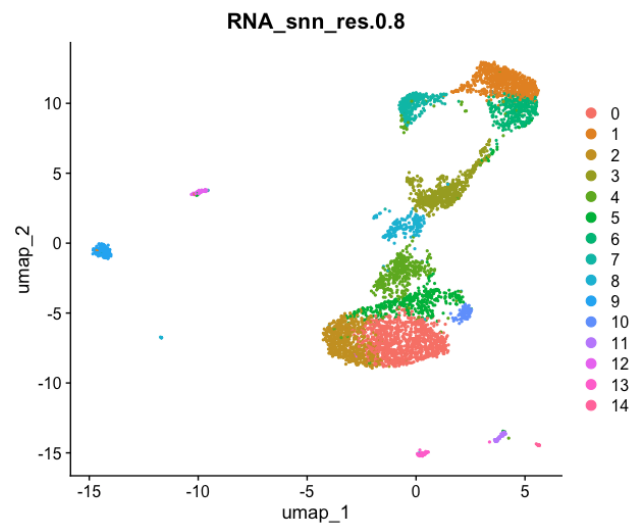
Dims = 10



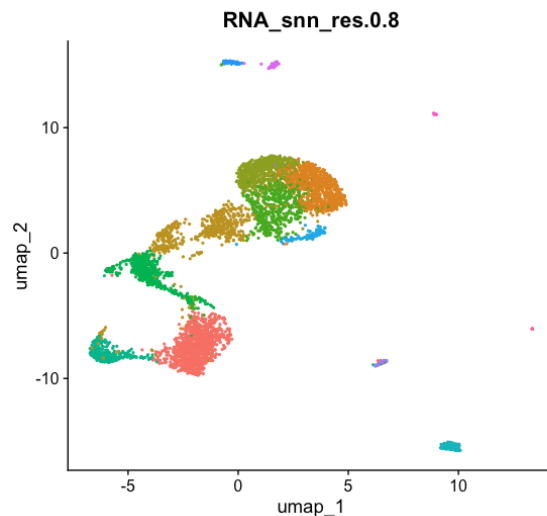
Dims = 20



Dims = 30



Dims = 40



The number of PCs changes both the separation of clusters and the number of clusters.

Clustering resolution

Next step after PCA is FindNeighbors() and FindClusters():

```
# Find neighbors:
seurat_data <- FindNeighbors(seurat_data, dims = 1:30)

# Clustering with resolution 0.8:
seurat_data <- FindClusters(seurat_data, resolution = 0.8)

# Test a range of resolutions
resolutions <- c(0.6, 0.7, 0.8, 0.9, 1)
seurat_data <- FindClusters(seurat_data,
                           resolution = resolutions
)
```

FindNeighbors() builds a **network of similarity**. It makes a neighbor graph that connects cells that are the most similar to each other.

FindClusters() takes the neighbor network from FindNeighbors() and looks for **groups that are more connected to each other** than the rest. It uses a community detection algorithm (Louvain or Leiden) to **group cells into clusters**.

You should **test multiple resolutions** with FindClusters

Clustering resolution

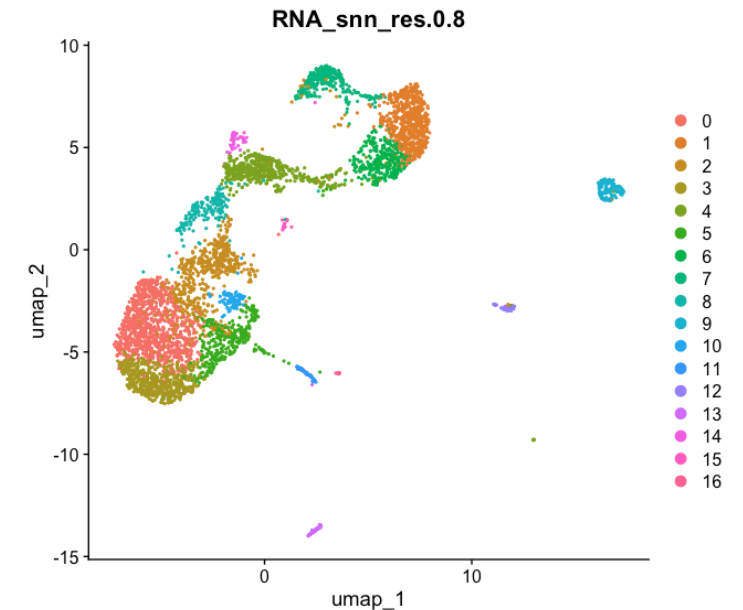
Next step is **RunUMAP**

```
seurat_data <- RunUMAP(seurat_data, dims = 1:20)
```

RunUMAP() takes the neighbor graph from FindNeighbors() and places it in a 2D plot.
It doesn't create clusters. it just visualizes them.

It's a nonlinear algorithm that tries to **preserve local neighborhoods** rather than exact distances. The positions of cells reflect **local similarity**, but distances are **not exact measures** of overall expression difference.

Thus, clusters can **appear farther or closer** than they really are in PCA space.

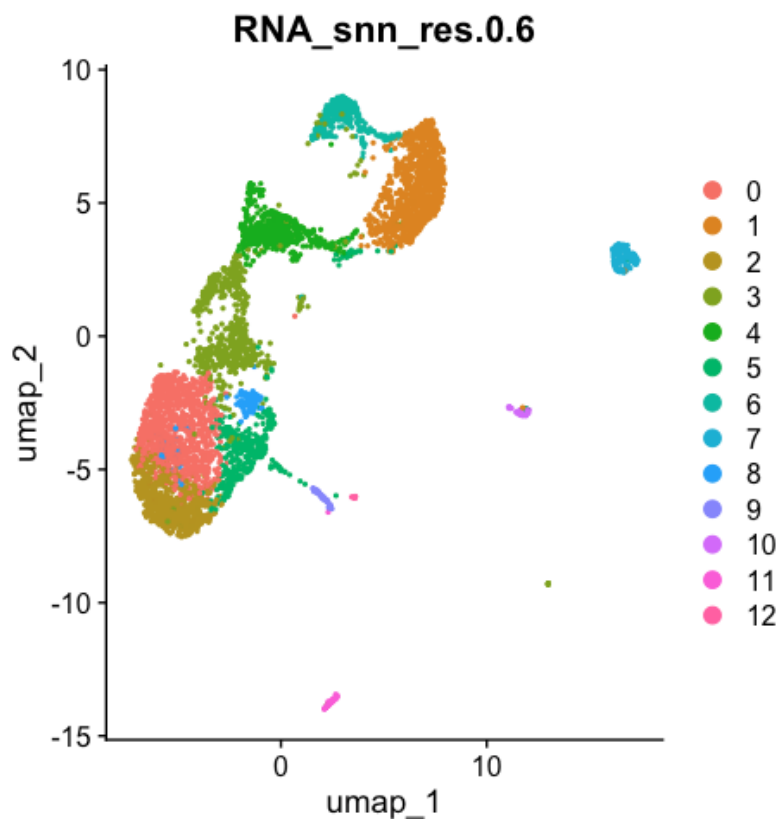


Clustering resolution

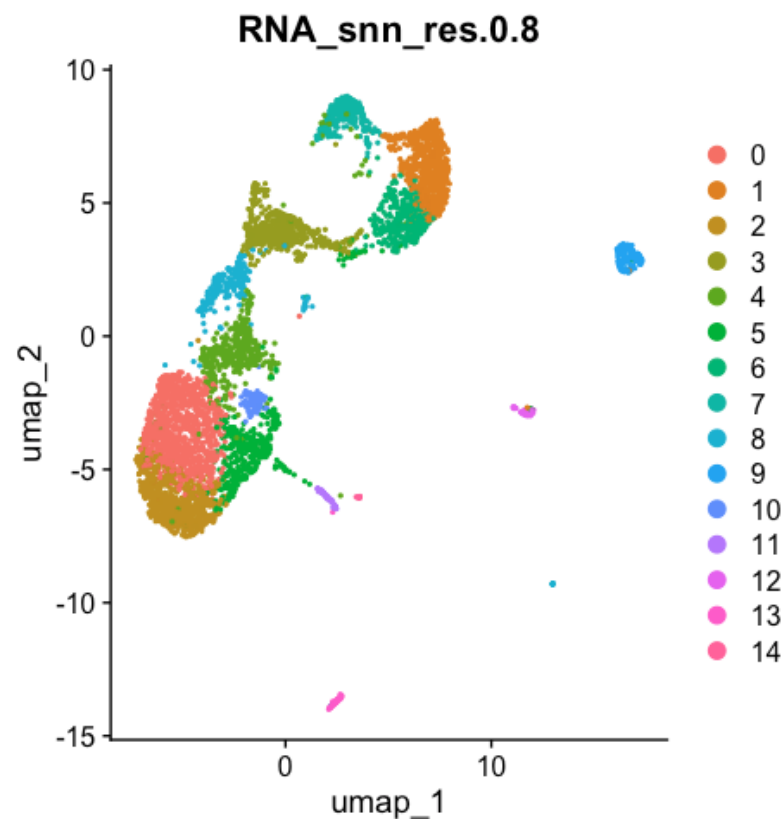
You can directly **affect the number of clusters** in your dataset by changing the resolution.

```
DimPlot(seurat_data, group.by = c("RNA_snn_res.0.6", "RNA_snn_res.0.8", "RNA_snn_res.1"), reduction = "umap")
```

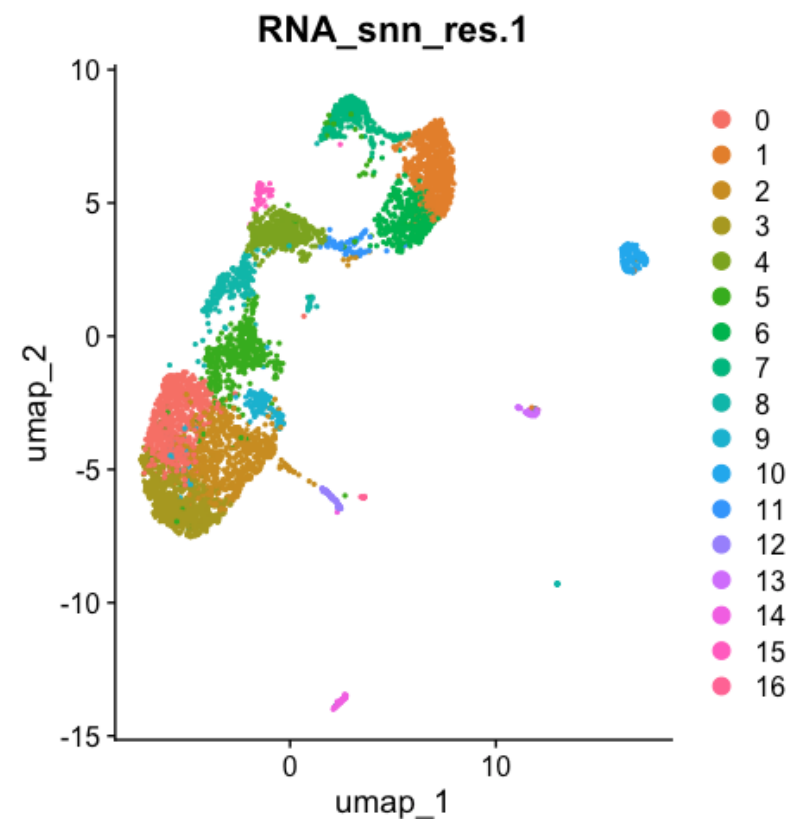
Resolution 0.6 gives you 12 clusters



Resolution 0.8 gives you 14 clusters



Resolution 1 gives you 16 clusters



Clustering resolution

Clustree can help you visualize how clusters **combine or divide** between resolutions

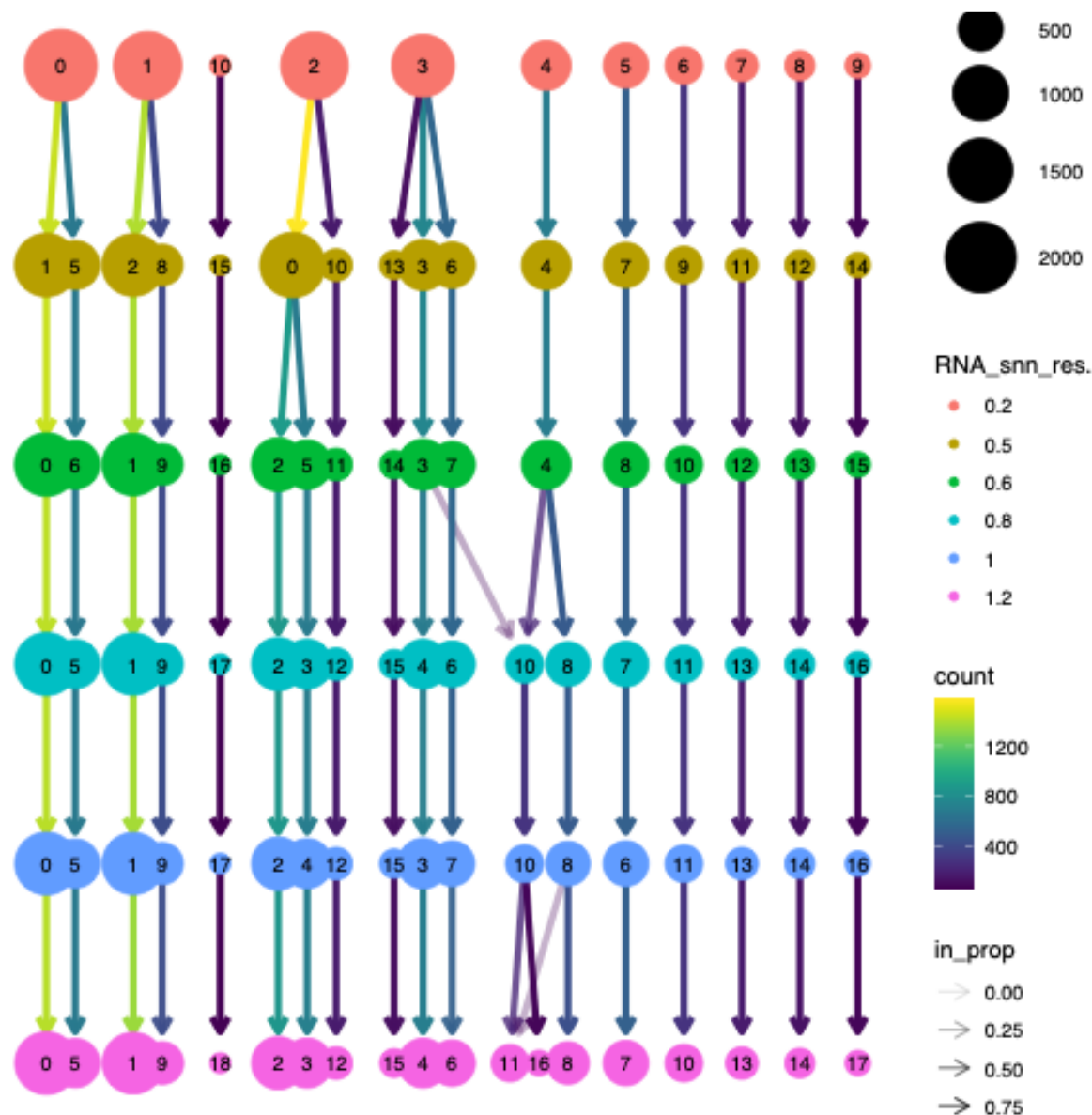
```
clustree(seurat_data, prefix = "RNA_snn_res.")
```

If a cluster splits into two new populations:

More believable biologically. Suggests that the original cluster was heterogeneous

If cells redistribute into already existing clusters:

Less informative biologically. Might reflect noise or cells that are borderline between clusters.



Clustering resolution

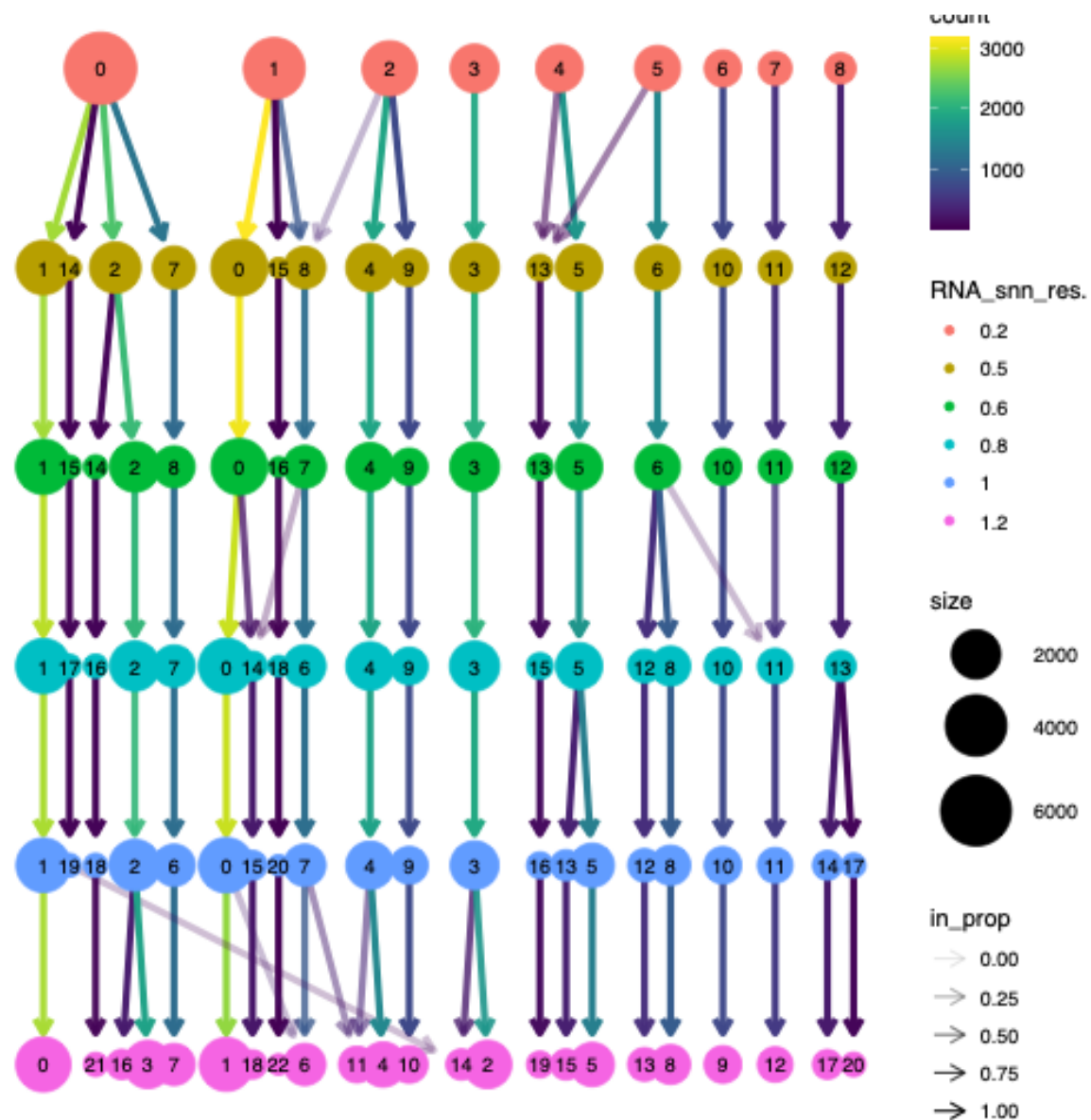
FindClusters() prefers clusters of similar size, even if that is not biologically true. Thus, it might be hard to divide small clusters.

It's better to overcluster and have clusters distribute into two new clusters. You can always merge these clusters later, if need be.

Cell number impacts clustering! The more cells you have, the easier it will be to get more clusters.

You don't have to use the same resolution in different samples! Different samples will likely need different resolutions, especially if the cell number is different!

Another example...



Clustering resolution

Signs you picked too few clusters:

- Known marker genes fail to segregate by cluster.
- Clusters are broad and heterogeneous.

Signs you picked too many clusters:

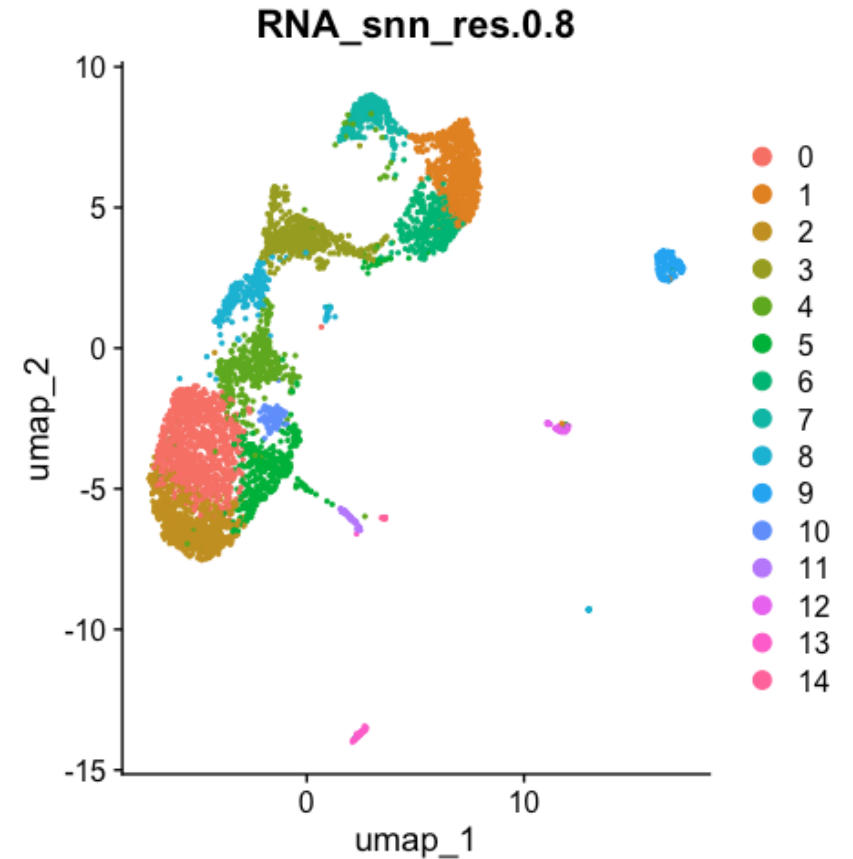
- Many tiny clusters that lack biological rationale.
- Marker genes do not match clusters.
- Clusters appear unstable across different random seeds.

Cell type identification

How do we find out what cell types cluster 1-16 are?

If possible, use a pre-existing dataset for initial cell type identification:

- Kristen has a repository of cell type references on the Alpine server for different organs and species:
 - `/p/active/Anschutz_BDC/resources/single_cell_references`



Cell type identification

Load the reference and use it to match your Seurat clusters to reference cell types

```
library(clustifyr)

ref_mat <- read.csv(file.path(save_dir, "clustifyr_mouse_reference.csv"),
                    header = TRUE, row.names = 1)

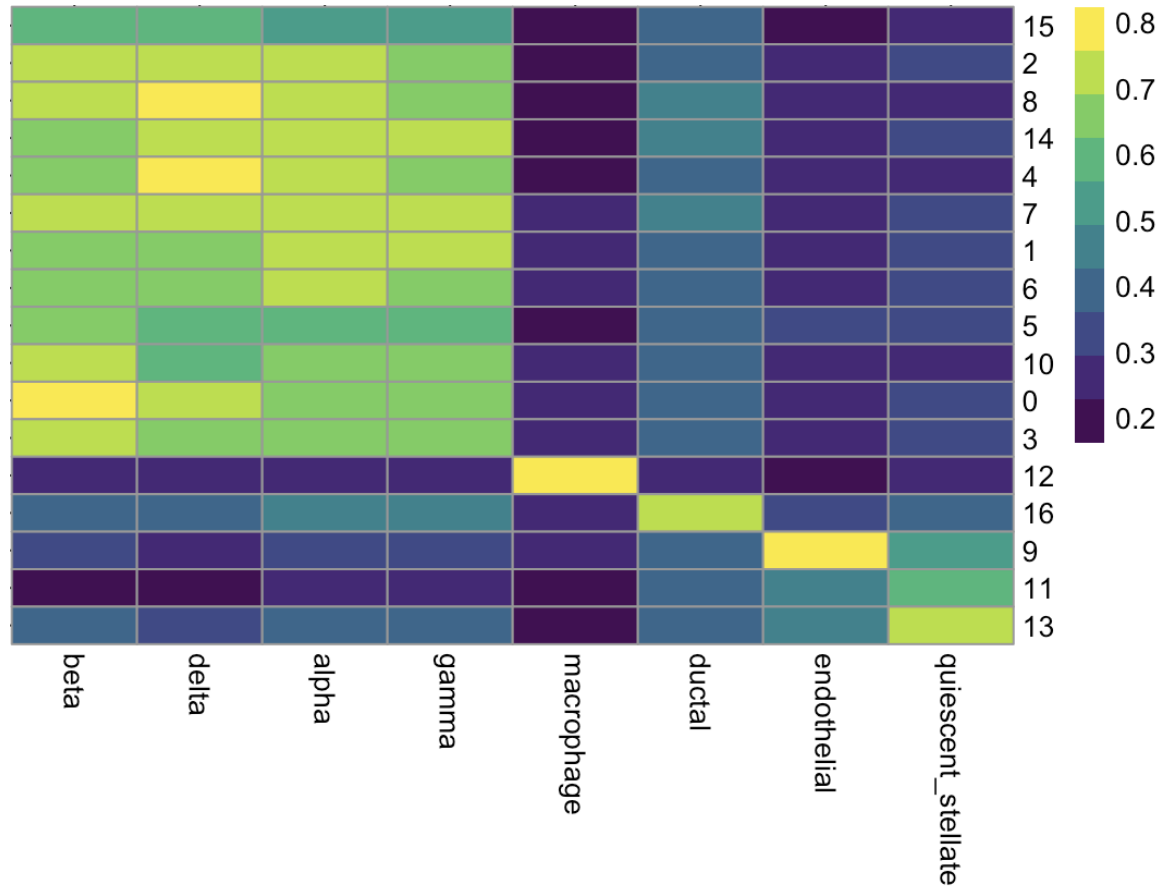
cluster_res <- name_clusters(seurat_data, ref_mat,
                             save_dir = save_dir,
                             save_name = "baron_celltype", ADT = FALSE,
                             assay = "RNA",
                             nfeatures = 1000, clusters = "RNA_cluster",
                             plot_type = "rna.umap")

seurat_data <- cluster_res$object
```

The reference is a gene expression matrix where rows are genes and columns are cell types. It contains average expression per gene per cell type.

name_clusters() is the Clustifyr function that matches your Seurat clusters to reference cell types.

Cell type identification



The Baron 2016 reference does a pretty good job at naming cell types

Other references are not quite as impressive...

Cell type identification

Make a table with the reference cell types and your clusters

```
> table(seurat_data$RNA_baron_celltype, seurat_data$RNA_cluster)
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
alpha	0	686	0	0	0	0	321	0	255	0	0	0	0	0	0	0	0	0	0
beta	707	0	614	566	0	0	0	261	0	0	0	88	0	0	0	0	0	0	0
delta	0	0	0	0	432	336	0	0	0	172	0	0	0	0	0	41	16	0	7
ductal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0
endothelial	0	0	0	0	0	0	0	0	0	0	129	0	0	0	0	0	0	0	0
macrophage	0	0	0	0	0	0	0	0	0	0	0	0	0	56	0	0	0	0	0
quiescent_stellate	0	0	0	0	0	0	0	0	0	0	0	0	63	0	52	0	0	0	0

How does clustering **resolution** change cell type identification?

```
table(seurat_data$RNA_baron_celltype, seurat_data$RNA_snn_res.0.8)
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
alpha	0	664	0	0	4	0	304	289	0	0	0	0	1	0	0	0	0
beta	942	0	554	545	1	104	0	0	8	0	82	0	0	0	0	0	0
delta	8	1	18	0	531	218	1	1	168	0	2	0	0	1	41	14	0
ductal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12
endothelial	0	0	0	0	0	0	0	0	0	129	0	0	0	0	0	0	0
macrophage	0	0	0	0	0	0	0	0	0	0	0	0	56	0	0	0	0
quiescent_stellate	0	0	0	0	0	0	0	0	0	0	0	63	0	52	0	0	0

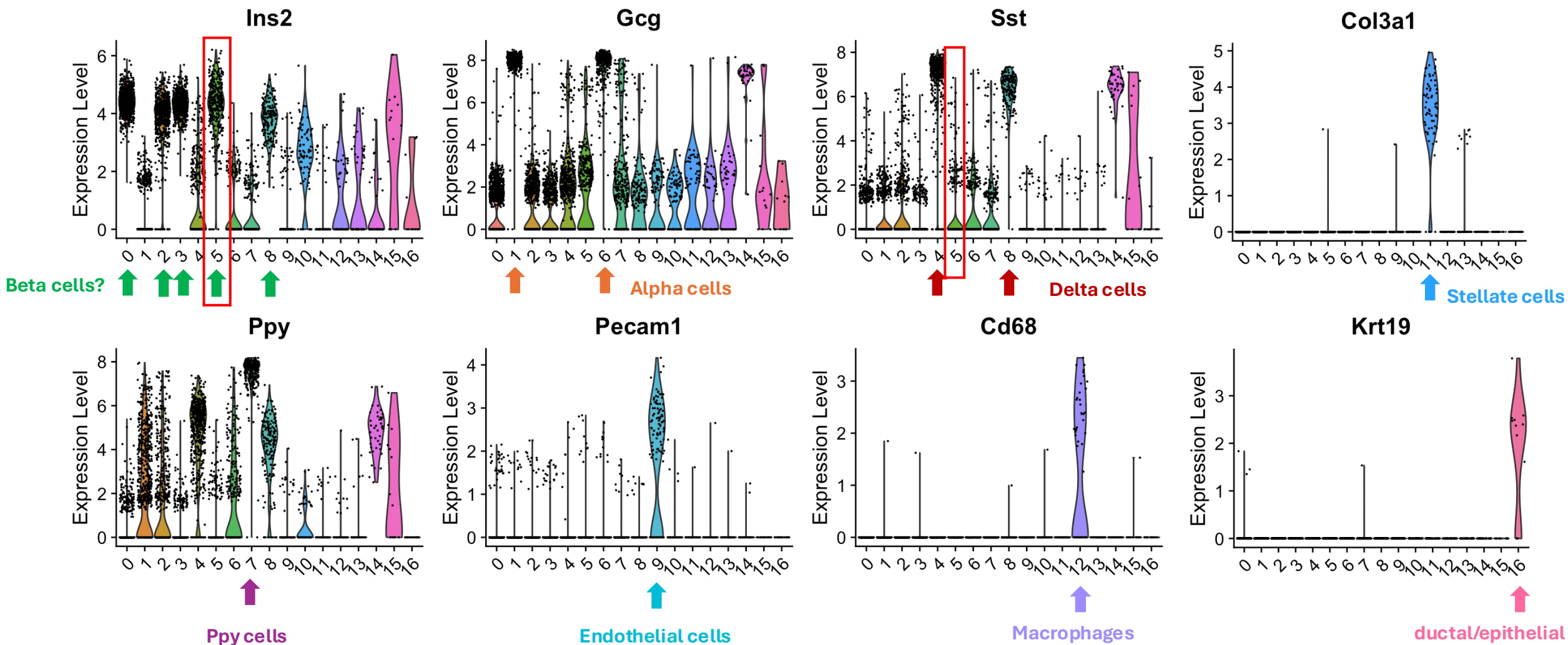
In general, the beta and delta cells seem to be the most transcriptionally similar, and the most difficult to separate.

What is cluster 5?

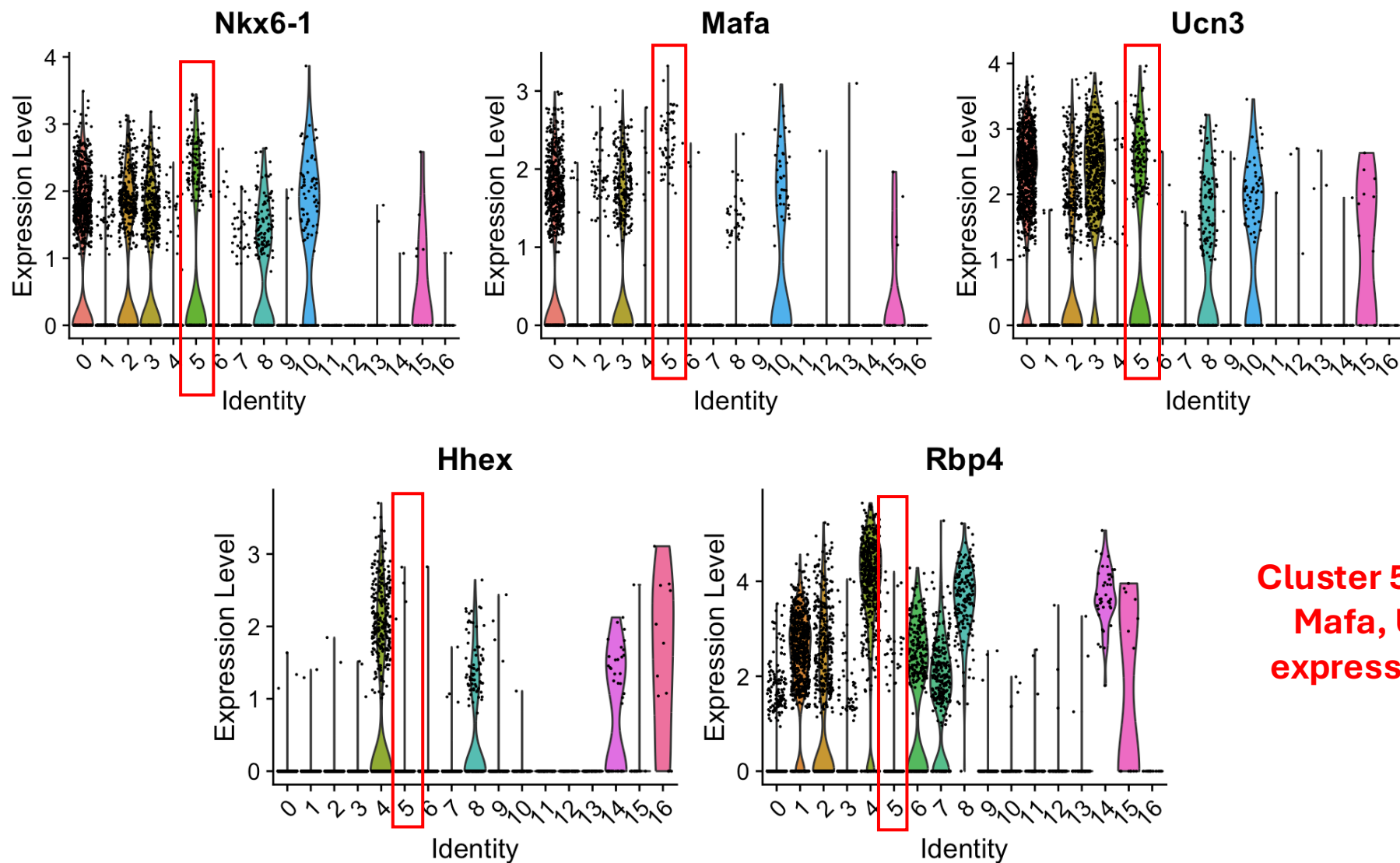
Cell type identification

Cluster 5 expresses insulin
but not somatostatin

Use gene expression of known markers to help identify the difficult populations



Cell type identification



Cluster 5 is expressing Nkx6-1, Mafa, Ucn3, but is missing expression of Hhex and Rbp4

Cell type identification

```
> table(seurat_data$RNA_cluster, seurat_data$RNA_snn_res.0.8)
```

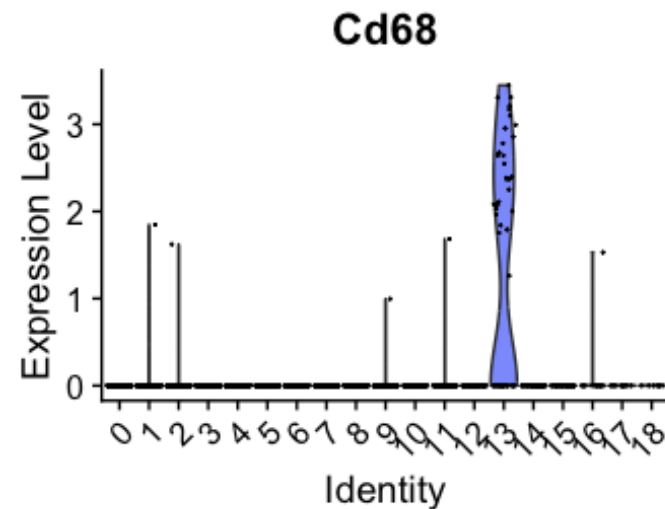
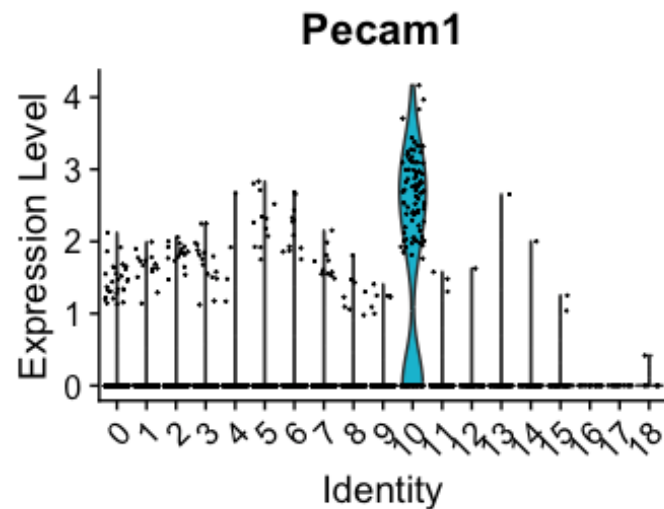
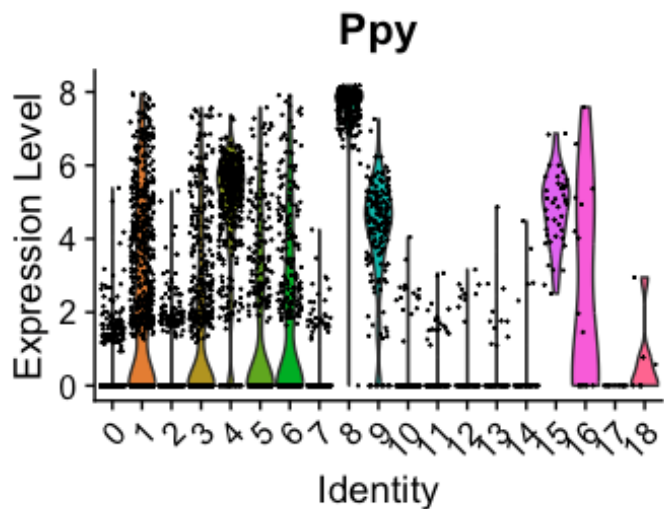
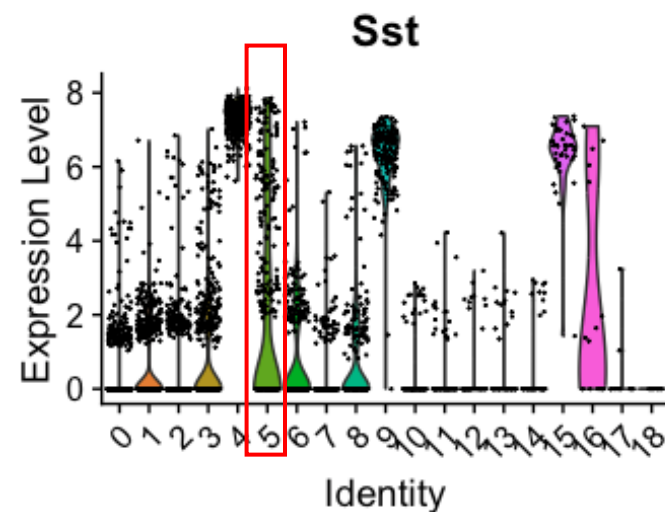
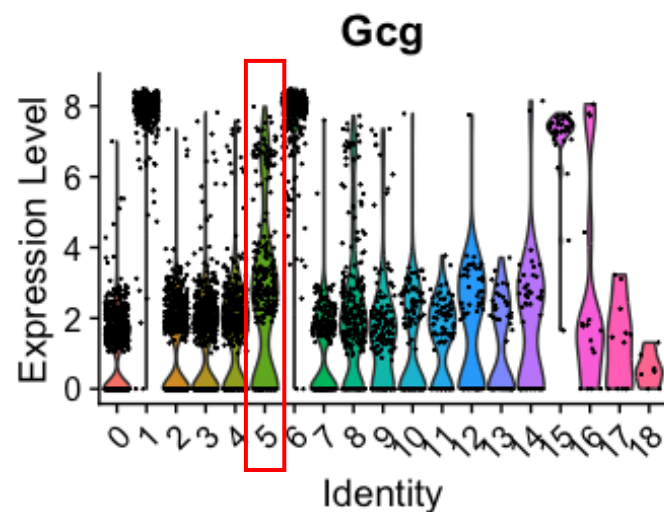
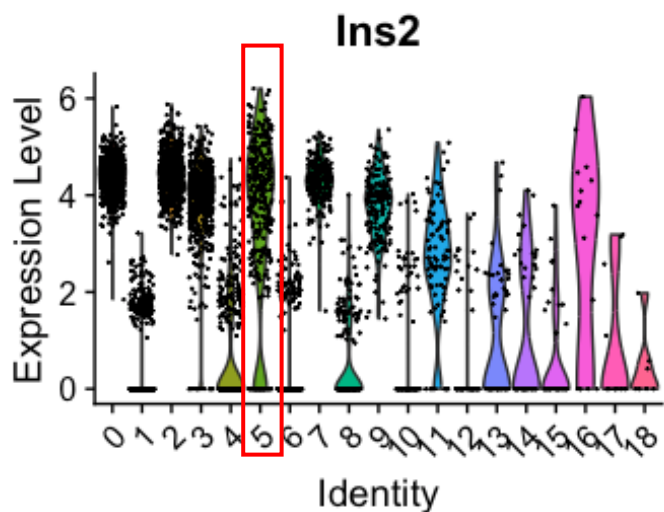
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	434	0	0	272	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	646	0	0	1	0	5	33	0	0	0	0	1	0	0	0	0
10	0	0	0	0	0	0	0	0	0	129	0	0	0	0	0	0	0
11	6	0	0	0	0	0	0	0	0	0	82	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	63	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0	56	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	0	52	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	41	0	0
16	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	13	0
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12
18	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0
2	225	0	11	272	0	104	0	0	2	0	0	0	0	0	0	0	0
3	27	0	532	1	1	0	0	0	5	0	0	0	0	0	0	0	0
4	0	0	0	0	432	0	0	0	0	0	0	0	0	0	0	0	0
5	7	0	18	0	89	218	1	0	0	0	2	0	0	0	0	1	0
6	0	18	0	0	3	0	299	1	0	0	0	0	0	0	0	0	0
7	250	0	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	255	0	0	0	0	0	0	0	0	0
9	0	0	0	0	3	0	0	0	168	0	0	0	0	1	0	0	0

Cluster 5 in the 0.8 resolution is made up of cluster 2 and 5 from the original clustering

Cluster 5 was called as delta in the original clustering

Cell type identification

Cluster 5 was probably under-clustered in the original clustering

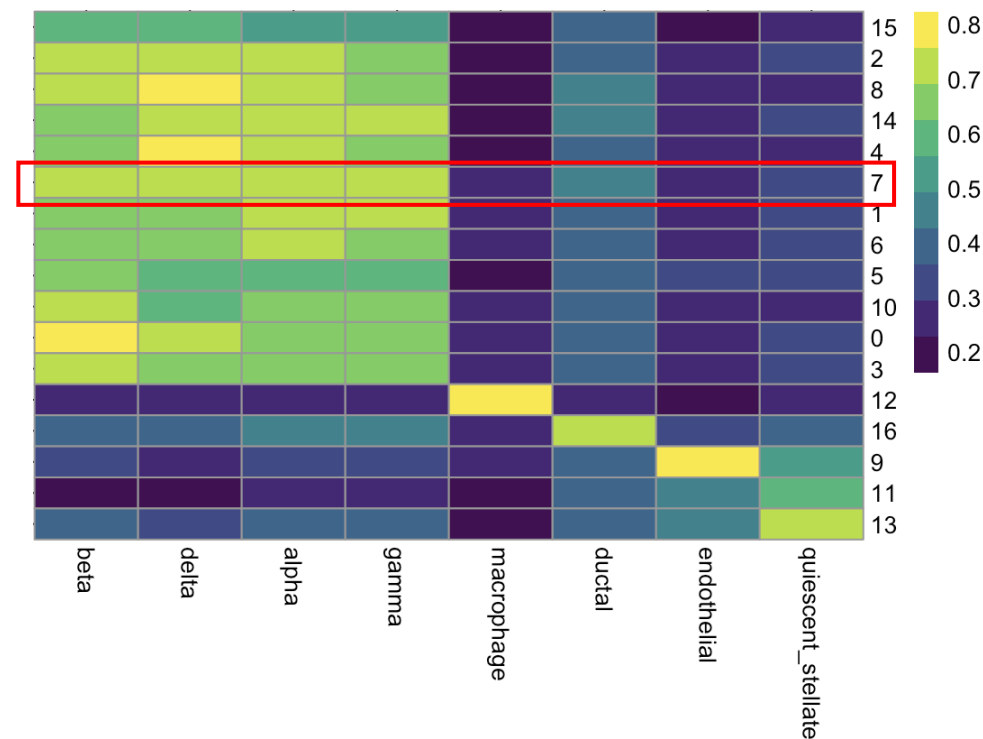


Cell type identification

What is **cluster 7**?

```
table(seurat_data$RNA_barcode_celltype, seurat_data$RNA_snn_res.0.8)
```

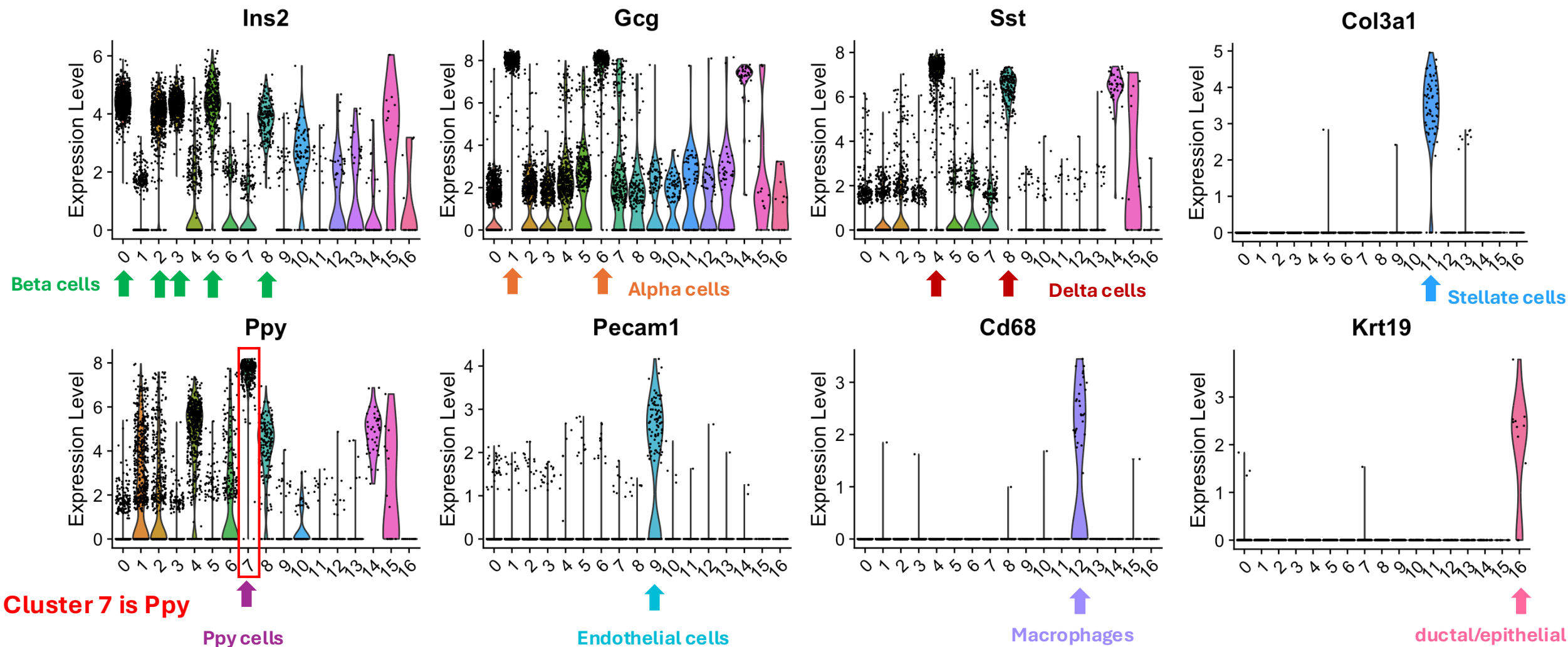
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
alpha	0	664	0	0	4	0	304	289	0	0	0	0	1	0	0	0	0
beta	942	0	554	545	1	104	0	0	8	0	82	0	0	0	0	0	0
delta	8	1	18	0	531	218	1	1	168	0	2	0	0	1	41	14	0
ductal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12
endothelial	0	0	0	0	0	0	0	0	0	129	0	0	0	0	0	0	0
macrophage	0	0	0	0	0	0	0	0	0	0	0	0	56	0	0	0	0
quiescent_stellate	0	0	0	0	0	0	0	0	0	0	0	63	0	52	0	0	0



It looks like alpha cells in the table,
but the gene expression doesn't seem
distinct in the heat map

Cell type identification

Use gene expression of known markers to help identify the difficult populations



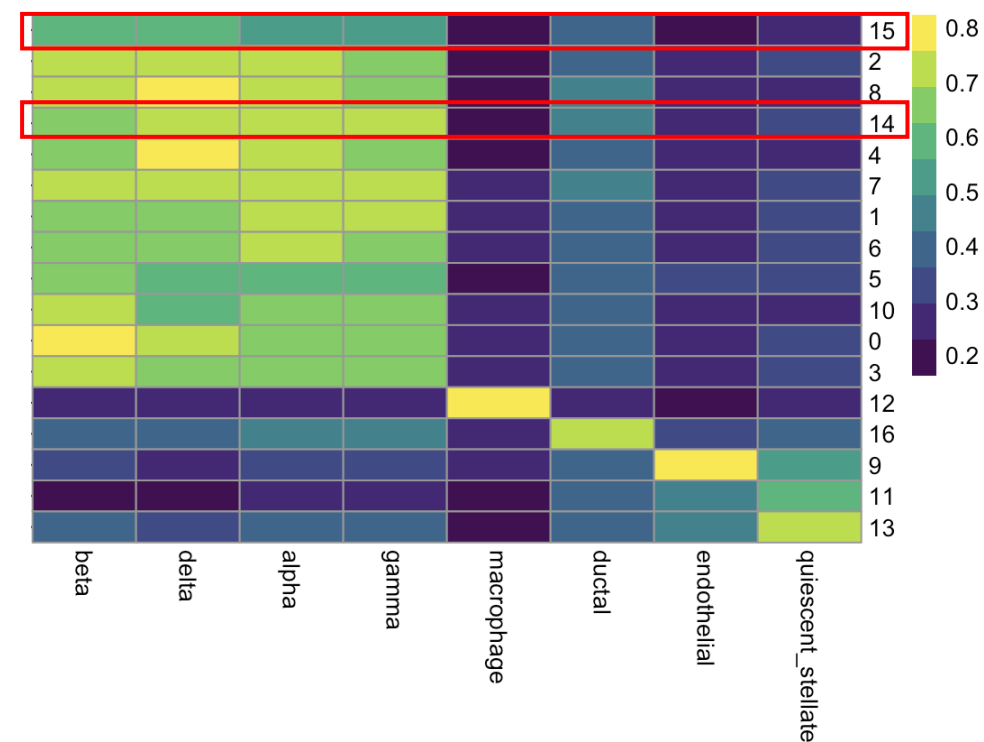
A cell type not included in the reference

Cell type identification

What are **cluster 14 and 15**?

```
table(seurat_data$RNA_barcode_celltype, seurat_data$RNA_snn_res.0.8)
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
alpha	0	664	0	0	4	0	304	289	0	0	0	0	1	0	0	0	0
beta	942	0	554	545	1	104	0	0	8	0	82	0	0	0	0	0	0
delta	8	1	18	0	531	218	1	1	168	0	2	0	0	1	41	14	0
ductal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12
endothelial	0	0	0	0	0	0	0	0	0	129	0	0	0	0	0	0	0
macrophage	0	0	0	0	0	0	0	0	0	0	0	0	56	0	0	0	0
quiescent_stellate	0	0	0	0	0	0	0	0	0	0	0	63	0	52	0	0	0

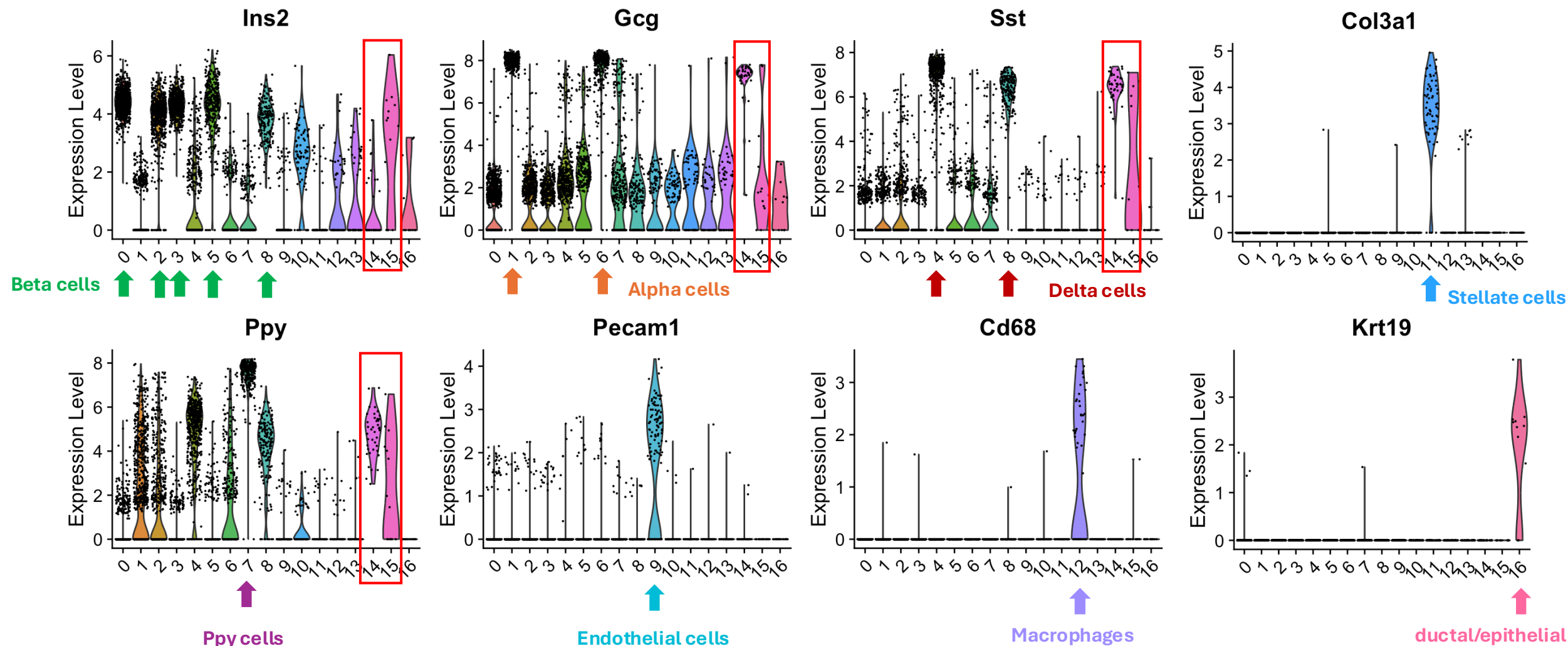


They look like delta cells in the table,
but their gene expression is not
distinct in the heat map

Cell type identification

Use gene expression of known markers to help identify the difficult populations

Cluster 14 and 15 are polyhormonal?



Cell type identification

Is the cluster truly polyhormonal?

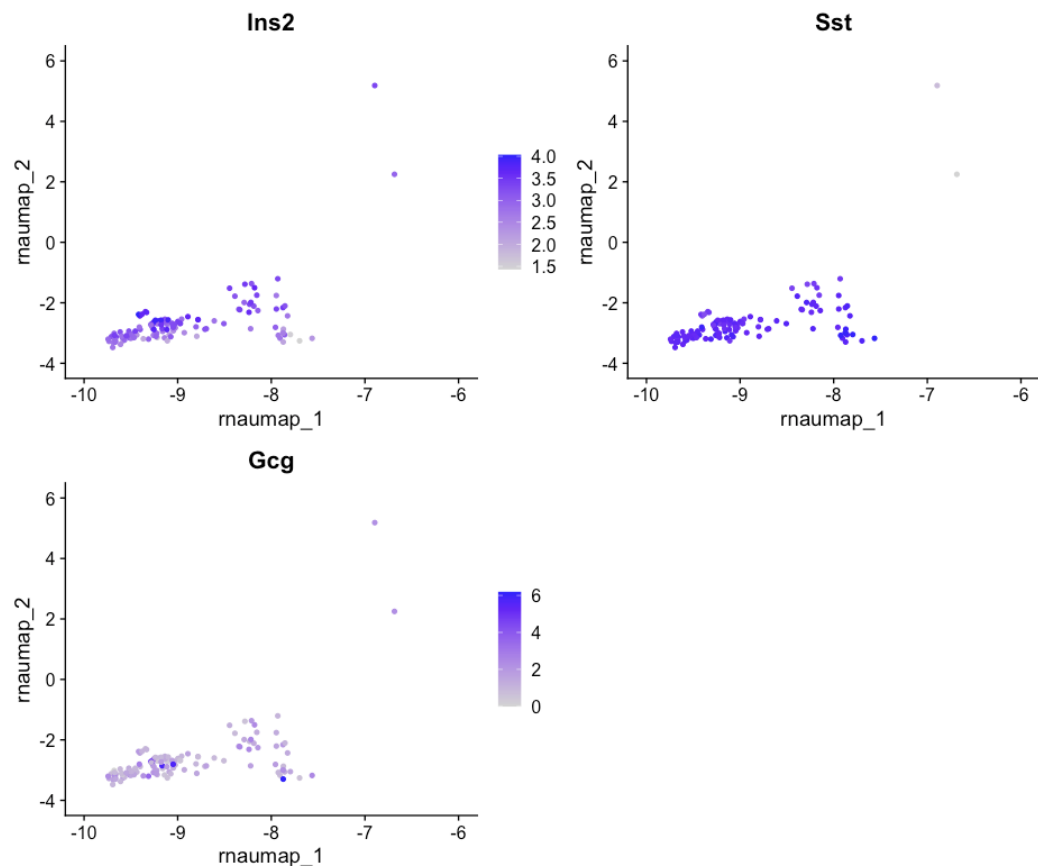
Subset the clusters you're interested in, and make plots showing hormone expression

```
FeaturePlot(pop14, c("Ins2", "Gcg", "Sst"))
```

```
FeaturePlot(pop15, c("Ins2", "Gcg", "Sst"))
```

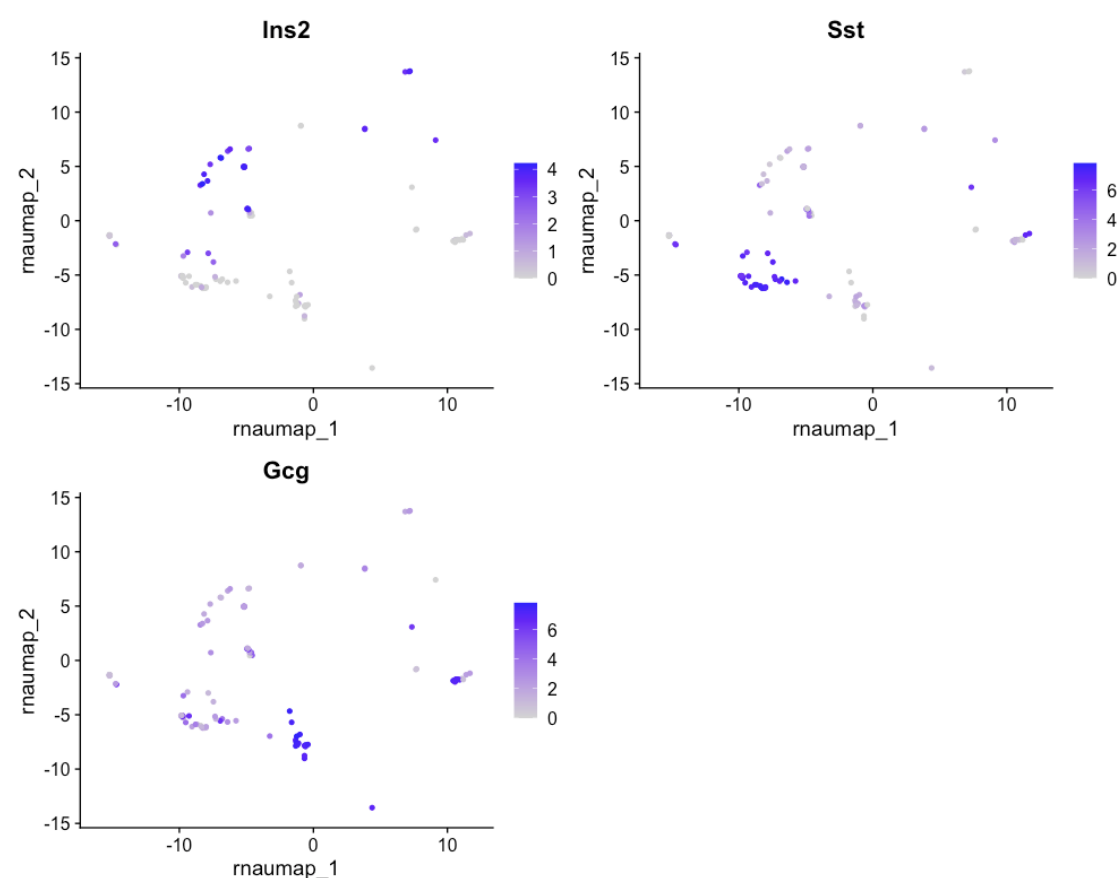
Polyhormonal

Is it a homogeneous population of cells that express multiple hormones?



Under-clustered

Is it a heterogeneous population of cells that should have been separated into multiple clusters?



Cell type identification

It's also a good idea to look at the top genes defining each population:

Use FindAllMarkers() to find genes differentially expressed in each cluster compared to all other cells:

```
markers <- FindAllMarkers(seurat_data)
```

Use FindMarkers() to find genes differentially expressed between two specific populations:

```
markers <- FindMarkers(seurat_data, ident.1 = "1", ident.2 = "2")
```

Be careful not to over conclude: You're likely always going to get differentially expressed genes between clusters, even if those clusters are pretty much the same cells. Especially if one cluster has a large number of cells, and the other cluster is small.

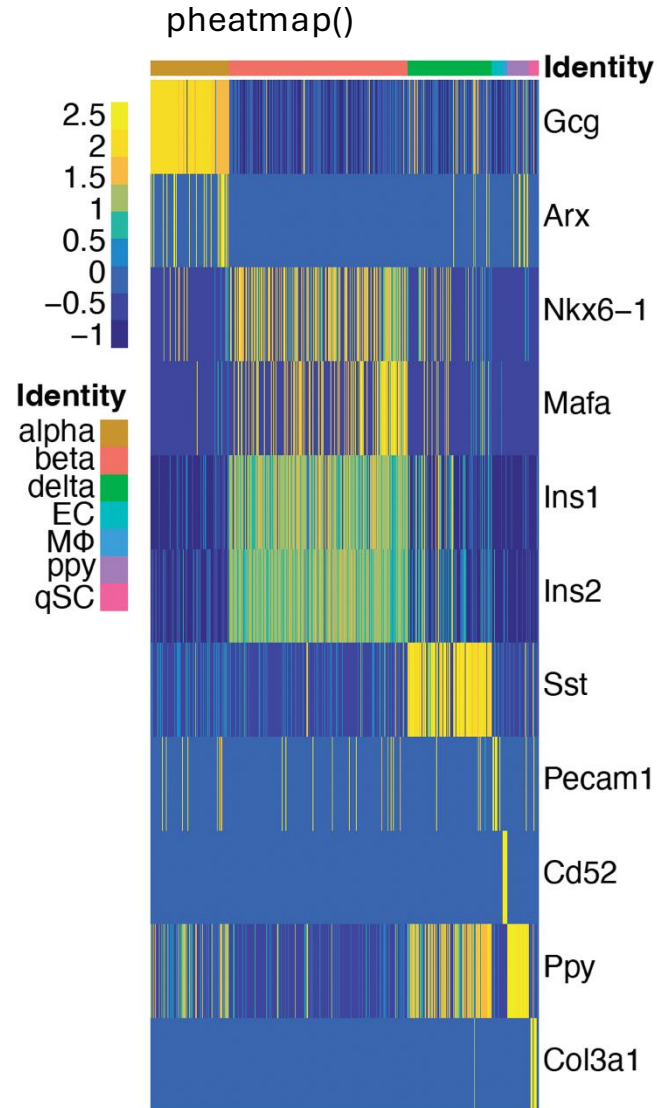
top_genes_per_cluster

cluster	gene
0	Prss53
0	G6pc2
0	Ins1
0	Ero1lb
0	Ucn3
1	Gcg
1	Ttr
1	Mafb
1	Pappa2
1	Gpx3
2	Chga
2	Cd79a
2	Txnip
2	Iapp
2	Scg3
3	Ftl1
3	Atf5
3	Nupr1
3	Cd63
3	Ppp1r1a

Top 5 genes expressed in cluster 0 (beta), compared to all other cells

Cell type identification

It's best practice to include a heat map showing marker expression of the different cell types you have identified:



Cluster with multiple replicates

Cluster and name cells in individual Seurat objects before merging them!

Order of sample merging:

- 1) Cluster and name cells in individual replicates/samples
- 2) Merge Seurat objects
- 3) Scale and find variable features again
- 4) Batch correct if needed
- 5) Cluster and name cells again
- 6) Check that the new naming is concordant with the original naming

Clusters driven by technical artifacts, such as high mitochondrial gene expression or low UMI counts, should be removed from each individual Seurat object before merging.

Batch correction

We perform batch correction to remove technical artifacts and enable meaningful comparisons.

Harmony and **fastMNN** are commonly used correction tools for single-cell RNA-seq analysis

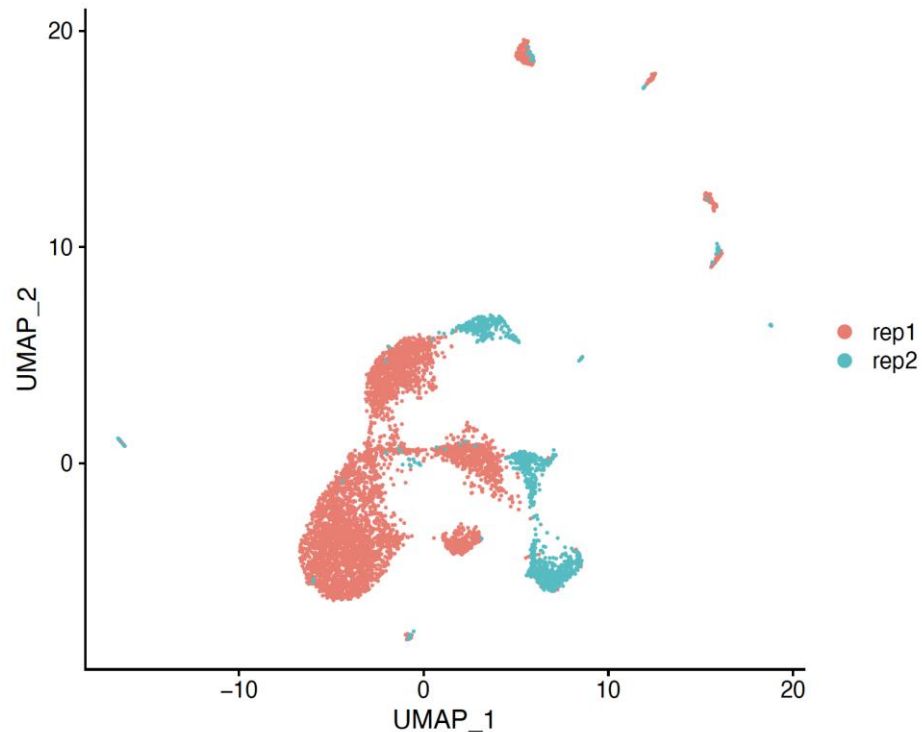
Goal: remove batch without removing biological structure.

Harmony **does not affect gene expression**, but it **affects clustering**.

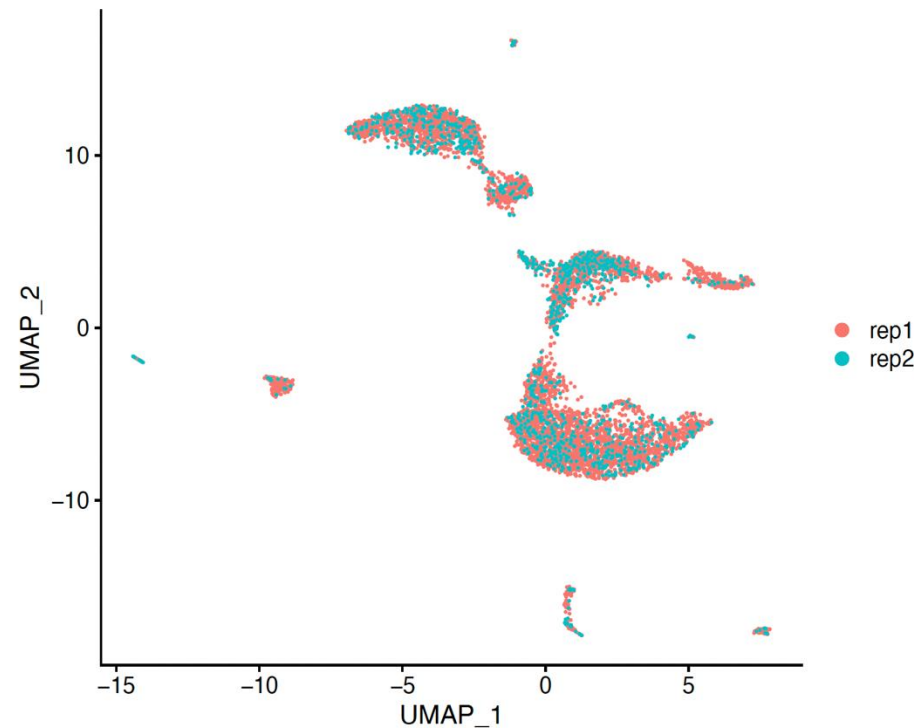
Overall, Harmony favors conservation of biological signal over integration.

Try multiple different tools to find the best integration!

Before batch correction

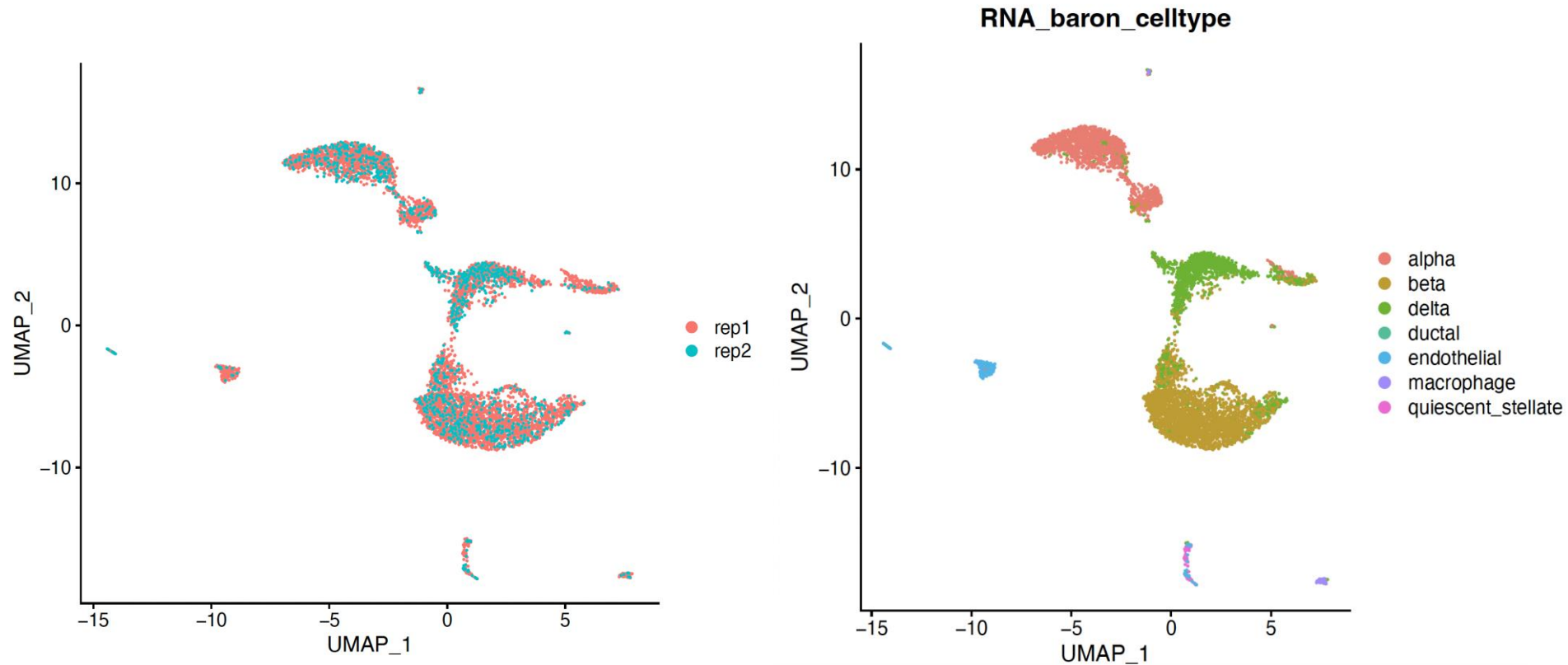


After batch correction



Batch correction

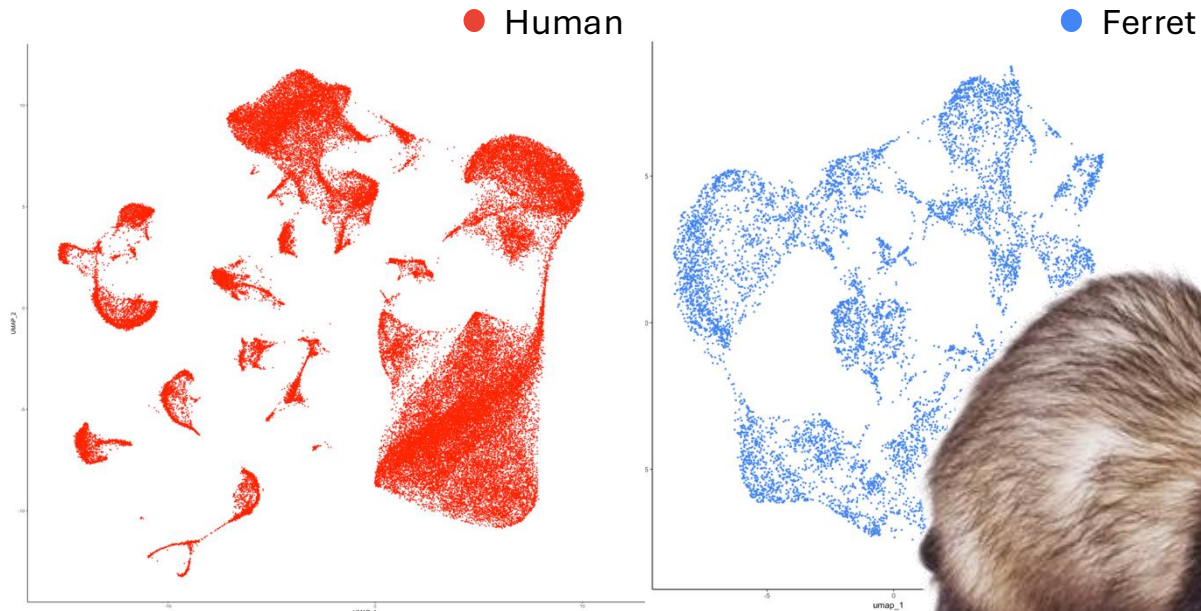
Make sure to check that the cell types are overlapping correctly
(you don't want to see alpha cells from replicate 1 overlapping with beta cells from replicate 2)



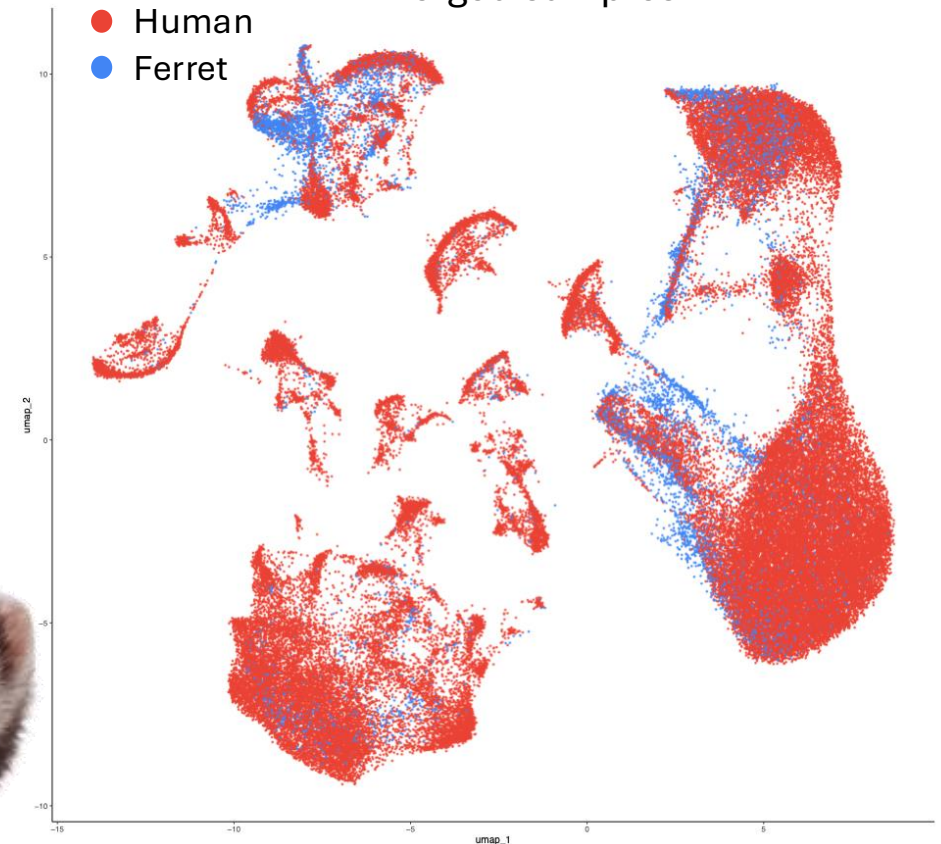
Batch correction

Another option is using **Seurat's batch correction tool** (IntegrateLayers)

This is a very **strong** tool, that will force the integration.
It favors integration over conservation of biological signal



Seurat batch-corrected merged samples



If you're **forcing** very different cells to cluster together
(in this case, islet cells from ferret and human), you can
use Seurat's **much more powerful** batch correction.

Note: benchmarking studies can help you determine
which tool is best for your type of study.

Is anything real??

Thousands of decisions go into a single-cell RNA-seq analysis.

You *will* make mistakes.

The key question is:

Can you still trust your conclusions?

What happens if you change some of your choices?

- 20 vs. 25 dimensions
- Resolution 0.8 vs. 0.9
- Calling an unclear beta/delta cluster “beta” vs. “delta”

If your conclusions **change**, they may not be robust.

If they **remain consistent**, that’s when you can have confidence in them.

Take-home messages

- Try out different number of PCs (dims).
- Try out different resolutions.
- If possible, use a reference dataset to assign cell types to your different resolutions.
 - Make sure the reference dataset is not missing cell types!
- Check which resolution best separate the cell types in your dataset. For this, use:
 - Gene expression of known marker genes.
 - Differentially expressed genes between clusters.
 - Check if clusters are homogeneous or heterogeneous.
- Perform clustering and cell type identification on all your samples/replicates individually.
 - Then merge, rerun `ScaleData()` and `FindVariableFeatures()`.
 - Batch correct and make sure the cell types remain the same.
 - Cluster again: `RunPCA()`, `FindNeighbors()`, `FindClusters()`.